Improving RAG Answer Quality by Learning to Balance Vector and Graph Retrieval

by

William Barrett

This thesis has been submitted in partial fulfillment for the degree of Master of Science in Artificial Intelligence

in the
Faculty of Engineering and Science
Department of Computer Science

March 2026

Declaration of Authorship

This report, Improving RAG Answer Quality by Learning to Balance Vector and Graph Retrieval, is submitted in partial fulfillment of the requirements of Master of Science in Artificial Intelligence at Munster Technological University Cork. I, William Barrett, declare that this thesis titled, Improving RAG Answer Quality by Learning to Balance Vector and Graph Retrievaland the work represents substantially the result of my own work except where explicitly indicated in the text. This report may be freely copied and distributed provided the source is explicitly acknowledged. I confirm that:

- This work was done wholly or mainly while in candidature Master of Science in Artificial Intelligence at Munster Technological University Cork.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at Munster Technological University Cork or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this project report is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Signed: _____

Date: _____

MUNSTER TECHNOLOGICAL UNIVERSITY CORK

Abstract

Faculty of Engineering and Science
Department of Computer Science

Master of Science

by William Barrett

Retrieval-Augmented Generation (RAG) dynamically integrates external knowledge into large language model-based (LLM) systems. LLMs perform well for general tasks but hallucinate when met with information they are not trained on. RAG is a proven solution to this but only performs well in specific chunk-based lookups. RAG fails to identify thematic data across an entire corpus. In 2024 Microsoft introduced a graph-based approach named GraphRAG to address this. GraphRAG prebuilds a graph model by identifying entities, communities, and relationships across an entire corpus. Most current research focuses on the integration of vector-based or graph-based RAG in isolation. In research where both are combined, the approach is a fixed equal weighting or selection of which to use. A current gap is that when both are combined, there is little to no research adjusting the split based on the incoming query. This means a question about how two entities relate will get treated the same as a fact lookup. The purpose of this research is to build and evaluate blending strategies to address this gap. These strategies will analyze incoming queries and determine the best means of blending the results of both vector and graph-based RAG. The contribution of this research will include an investigation into multiple classification and blending strategies, combined with an investigation into if adaptive routing outperforms individual retrieval strategies.

Contents

Declaration of Authorship **i**

Abstract **ii**

Abbreviations **iv**

1 **1**

1.1 Research Context 1

1.2 Research Aim 2

1.3 Research Objectives 2

Bibliography **3**

Abbreviations

LLM	**L**arge **L**anguage **M**odel
RAG	**R**etrieval-**A**ugmented **G**eneration
RRF	**R**ecipricol **R**ank **F**usion

Chapter 1

1.1 Research Context

Lewis et al. [1] introduced Retrieval-Augmented Generation (RAG) to enhance large language models (LLMs) with external knowledge through vector based retrieval. Later, Edge et al. [2] highlighted that vector RAG falls short when it comes to global or sense making queries and proposed a graph based approach called GraphRAG. GraphRAG allows the system to answer global queries like 'what are the main themes in this dataset?'. However, GraphRAG has its own limitations. Graph construction introduces an expensive upfront cost, and for simple query-focused summarization, vector RAG is far more efficient. This highlights the need for hybrid approaches to retrieving knowledge outside of parametric memory.

Xu et al. [3] tested this by deploying a hybrid model to production for six months. This hybrid approach was shown to reduce the median per-issue resolution time by 28.6%. However, their system relies on a fallback approach where they try graph search first and on failure it reverts to vector-based retrieval. There is no means to blend both sources and adjust the balance based on the query.

Jeong et al. [4] proposed Adaptive RAG which trains a classifier to classify the query and route between no retrieval, single-step, or multi-step retrieval. This outperformed single step retrieval and used half the compute time when performing multi-step approaches; this proves that classifying queries to pick the right strategy improves retrieval quality. However, this works purely against vector-based RAG and does not consider graph retrieval at all.

Guo et al. [5] used reinforcement learning to fine tune an LLM to route between vector and graph retrieval, but this requires fine tuning the LLM's weights which makes adoptability difficult.

Min et al. [6] combined vector and graph retrieval using RRF [7] and showed a 15% improvement over vector-only baselines. However, their system runs both retrieval strategies for every query with a fixed equal weighting. There is no mechanism to skip graph or vector retrieval when the query doesn't need it.

No current work uses an adaptive balancing approach to blend vector and graph-based retrieval. This research investigates strategies to address this gap and allow better question answering across query types by weighting the retrieval type based on the incoming query.

1.2 Research Aim

The aim of this research is to investigate if adaptive, classification-based blending of vector and graph retrieval can outperform fixed strategies in RAG systems.

1.3 Research Objectives

1. To investigate if rule-based, LLM-based and model-based classifiers can determine retrieval weights between vector and graph to provide a lightweight alternative to fine tuning the LLM directly.

2. To determine if a weighted blending strategy can outperform fixed retrieval strategies that apply the same retrieval approach regardless of query type.

3. To identify which query characteristics should influence the balance between vector and graph retrieval.

4. To determine if a threshold exists to skip a retrieval type entirely, addressing the overhead of running both for every query.

Bibliography

[1] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive nlp tasks," in *Proceedings of the 34th International Conference on Neural Information Processing Systems*, 2020.

[2] D. Edge, H. Trinh, N. Cheng, J. Bradley, A. Chao, A. Mody, S. Truitt, D. Metropolitansky, R. O. Ness, and J. Larson, "From local to global: A graph rag approach to query-focused summarization," 2025. [Online]. Available: https://arxiv.org/abs/2404.16130

[3] Z. Xu, M. J. Cruz, M. Guevara, T. Wang, M. Deshpande, X. Wang, and Z. Li, "Retrieval-augmented generation with knowledge graphs for customer service question answering," in *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2024.

[4] S. Jeong, J. Baek, S. Cho, S. J. Hwang, and J. Park, "Adaptive-RAG: Learning to adapt retrieval-augmented large language models through question complexity," in *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, 2024.

[5] Y. Guo, M. Su, S. Guan, Z. Sun, X. Jin, J. Guo, and X. Cheng, "Routerag: Efficient retrieval-augmented generation from text and graph via reinforcement learning," 2025. [Online]. Available: https://arxiv.org/abs/2512.09487

[6] C. Min, S. Bansal, J. Pan, A. Keshavarzi, R. Mathew, and A. V. Kannan, "Towards practical graphrag: Efficient knowledge graph construction and hybrid retrieval at scale," 2025. [Online]. Available: https://arxiv.org/abs/2507.03226

[7] G. V. Cormack, C. L. A. Clarke, and S. Buettcher, "Reciprocal rank fusion outperforms condorcet and individual rank learning methods," in *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2009.